B05 — DCF valuation wo

Classification
Prepared
Business
Entity
Currency / units
asOf
Valuation date
Sheet guide
Colour convention

Limitations
Missing upstream assets

rkbook

TEST_ONLY — synthetic inputs; not commercial data
Creator-side P4 capability sample; NOT an Agent-driven delivery and NOT reviewed
finance-model-builder / dcf-model method (direct Skill method, not the comps-first /dcf command)
TOY Corp (fictional)
USD millions unless stated; per-share values in USD
2025-01-31
FY2024A actuals; five forecast years FY2025E–FY2029E; year-end discounting
Inputs (blue = hard-coded) → WACC → DCF → Sensitivity (3 grids × 25 live cells) → Checks
Blue = input, black = formula within the sheet, green = link to another sheet
No comps/precedents, no LBO, no three-statement model, no scenario-specific WACC inside the grids, no P5 review gate, no source licensing. Bear/Base/Bull change growth, margin, beta and terminal growth only. LBO/comps/Nike binary templates are MISSING_SOURCE_ASSET in the preserved archive; this workbook never claims to restore them.

Inputs — blue figures are hard-coded

Operating inputs	
Revenue FY2024A (USD m)	1,000.0
EBITDA margin FY2024A	20.0%
D&A % of revenue	4.0%
Capex % of revenue	5.0%
Change in NWC % of revenue change	2.0%
Tax rate	25.0%
Net debt (USD m)	150.0
Diluted shares (m)	100.0

Market inputs	
Risk-free rate	4.0%
Equity risk premium	5.0%
Pre-tax cost of debt	5.0%
Debt / total capital	30.0%

Scenario assumptions

Scenario	Revenue growth	EBITDA margin	Beta
Bear	3.0%	18.0%	1.30
Base	6.0%	20.0%	1.10
Bull	9.0%	22.0%	0.90

Active scenario (1 = Bear, 2 = Base, 3 = Bull)	2

Active assumptions — formulas	
Scenario name	Base
Revenue growth	6.0%
EBITDA margin	20.0%
Beta	1.10
Terminal growth	2.0%

Terminal growth
1.5%
2.0%
2.5%

WACC — CAPM for the active scenario

Cost of equity	
Risk-free rate	4.0%
Beta	1.10
Equity risk premium	5.0%
Cost of equity	9.5%
Cost of debt	
Pre-tax cost of debt	5.0%
Tax rate	25.0%
After-tax cost of debt	3.8%
Capital structure	
Debt / total capital	30.0%
Equity / total capital	70.0%
WACC	
WACC (active scenario)	7.8%

DCF — five-year explicit forecast with Gordon terminal value

USD millions	FY2024A	FY2025E	FY2026E
Revenue	1,000.0	1,060.0	1,123.6
Revenue growth		6.0%	6.0%
EBITDA		212.0	224.7
EBITDA margin		20.0%	20.0%
D&A		42.4	44.9
EBIT		169.6	179.8
Tax on EBIT		(42.4)	(44.9)
NOPAT		127.2	134.8
D&A add-back		42.4	44.9
Capex		(53.0)	(56.2)
Change in NWC		(1.2)	(1.3)
Free cash flow		**115.4**	**122.3**

Discounting			
Period (years)		1	2
Discount factor		0.928	0.861
PV of free cash flow		107.1	105.3

Terminal value	
Terminal growth (g)	2.0%
WACC	7.8%
Terminal-year FCF (FY2030E)	148.6
Terminal value (Gordon)	2,573.2
PV of terminal value	1,769.6

Valuation bridge	
Sum of PV of free cash flow	518.0
PV of terminal value	1,769.6
Enterprise value	**2,287.7**
Less: net debt	(150.0)
Equity value	**2,137.7**
Diluted shares (m)	100.0
Value per share (USD)	**21.38**

Model checks	
Five forecast years discounted	OK
Terminal growth below WACC	OK
No negative FCF in the forecast	OK
Value per share positive	OK

FY2027E	FY2028E	FY2029E
1,191.0	1,262.5	1,338.2
6.0%	6.0%	6.0%
238.2	252.5	267.6
20.0%	20.0%	20.0%
47.6	50.5	53.5
190.6	202.0	214.1
(47.6)	(50.5)	(53.5)
142.9	151.5	160.6
47.6	50.5	53.5
(59.6)	(63.1)	(66.9)
(1.3)	(1.4)	(1.5)
129.7	**137.4**	**145.7**

3	4	5
0.799	0.741	0.688
103.6	101.9	100.2

Sensitivity — live closed-form grids tied to the DCF

Each grid is a 5×5 live formula block; the centre cell of each grid must equal DCF!B36. The closed form is the algebraic equiva margin and year-end discounting.

Factor definitions	
Revenue FY2024A	1,000.0
Revenue growth (r)	6.0%
Base EBITDA margin	20.0%
D&A % revenue	4.0%
Capex % revenue	5.0%
NWC % of revenue change	2.0%
Tax rate	25.0%
Net debt	150.0
Diluted shares	100.0
Base WACC	7.8%
Base terminal growth	2.0%

Grid 1 — WACC (rows) × terminal growth (columns): value per share

Base factor K (base margin)	0.10887			
	1.5%	1.8%	2.0%	2.3%
6.8%	24.02	25.08	26.25	27.55
7.3%	21.78	22.64	23.58	24.62
7.8%	19.89	20.60	21.38	22.22
8.3%	18.28	18.88	19.52	20.22
8.8%	16.89	17.40	17.94	18.53

Grid 2 — WACC (rows) × EBITDA margin (columns): value per share

Margin per column	18.0%	19.0%	20.0%	21.0%
Factor K	0.09387	0.10137	0.10887	0.11637
6.8%	22.43	24.34	26.25	28.16
7.3%	20.13	21.85	23.58	25.31
7.8%	18.22	19.80	21.38	22.95
8.3%	16.63	18.08	19.52	20.97
8.8%	15.27	16.60	17.94	19.28

Grid 3 — terminal growth (rows) × EBITDA margin (columns): value per share

Base Σ q^t at base WACC

Margin per column	18.0%	19.0%	20.0%	21.0%
Factor K	0.09387	0.10137	0.10887	0.11637
1.5%	16.94	18.41	19.89	21.36
1.8%	17.56	19.08	20.60	22.12
2.0%	18.22	19.80	21.38	22.95

2.3%	18.95	20.59	22.22	23.86
2.5%	19.75	21.45	23.15	24.85

lent of the explicit DCF for constant growth, constant

2.5%
29.00
25.77
23.15
20.99
19.17

Σ q^t
4.8922
4.8245
4.7583
4.6935
4.6301

22.0%
0.12387
30.07
27.04
24.53
22.42
20.62

Σ q^t
4.8922
4.8245
4.7583
4.6935
4.6301

Σ q^t at base WACC
4.7583

22.0%
0.12387
22.83
23.65
24.53

25.49
26.55

Checks — tie-outs and integrity

Test	Result	Status
DCF value per share	21.38	
Grid 1 centre equals DCF	21.38	OK
Grid 2 centre equals DCF	21.38	OK
Grid 3 centre equals DCF	21.38	OK
WACC independent recompute	7.8%	OK
EV equals PV(FCF) + PV(TV)	2,287.7	OK
Equity equals EV minus net debt	2,137.7	OK
Scenario selector in range	2	OK
All DCF model checks OK	OK	OK
Sensitivity cells populated	75	OK
Sensitivity minimum positive	15.27	OK

Known limitations

TEST_ONLY synthetic inputs; no commercial data, licence or vendor access.

Grids vary one parameter pair around the active scenario and hold the others fixed; they are not probability weighting.

Year-end discounting only; no mid-year convention toggle in this sample.

No comps, precedents, LBO, three-statement model, football field or deck.

P5 review gate not implemented: this file is DRAFT and must not be distributed.